The Four Layers: Prompt, Context, Harness, Loop

General Agents, Chapter 7/12 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/four-layers

EN: The 4 layers are CONTAINERS, not steps -- nested inside each other, each responsible for a different unit of work. When something breaks, the first question is: which layer broke?
Roman Urdu: 4 layers CONTAINERS hain, steps nahi -- ek doosre ke andar nested, har ek alag unit-of-work ka zimmedar. Jab kuch toote, pehla sawal: kaunsi layer toti?

1. The Unit-of-Work Test

- Prompt = one model call. Context = the whole window. Harness = one beat (instruction to agent-goes-quiet). Loop = the whole run (nobody typing).

2. The Four Layers, One by One

- **Prompt** breaks when the task was understood, work is roughly right, but shape/length/voice is off.
- **Context** breaks when the answer is confident, fluent, factually wrong (often near some OTHER true fact). Harness is a curator (order, compression, dropping policy) -- not just a search box.
- **Harness** job: assemble context, call the model, run tools, return results, handle errors, enforce before the beat ends. 'Check passed' != 'check passing was ENOUGH' -- that's the Loop's job.
- **Loop** needs: Heartbeat (what starts it), Spine (state outside the model), Outside stops (success condition, limits, no-progress check, separate checker).

3. Human Gate + Using the Map

- Stops = safely FAIL; Gate = succeed WITH HELP. Guessing is dangerous because it's silent (a crash is loud).
- **Diagnostic table:** shape/tone wrong but task understood -> Prompt; confident-fluent-wrong -> Context; unproven success claim -> Harness; unchecked wrong answer propagates -> Loop/checker; never stops/stops too early/guesses -> Loop/stops-gate. This is a search order, not a verdict -- failures cross layers.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Unit-of-work test	Prompt=one call, Context=the window, Harness=one beat, Loop=the whole run.	Prompt=ek call, Context=window, Harness=ek beat, Loop=poori run.
Harness as curator	Order + compression + dropping policy decide what the model sees.	Order + compression + dropping policy decide karte model kya dekhe.
'Check passed' != 'enough'	A harness check passing doesn't prove the outcome was actually good -- that's the loop's job.	Harness check pass hona outcome achha hone ka proof nahi -- yeh loop ka kaam hai.
Human Gate	A planned pause that succeeds WITH help, unlike a stop that just fails safely.	Planned pause jo madad se succeed hota, stop ki tarah sirf safely fail nahi.

Quick Self-Check

Question	Answer
What's the unit-of-work test for telling the 4 layers apart?	Prompt = one model call, Context = the whole window, Harness = one beat, Loop = the whole run.
What's the key sign that Context (not Prompt) broke?	A confident, fluent answer that's factually wrong -- often close to some other true fact.
Why isn't 'the check passed' the same as 'the work was actually good'?	A beat can end for many reasons (timeout, error, model decision) -- a passing check is evidence, not proof; verifying that's enough is the loop's job.
What's the difference between a Stop and a Gate?	A Stop fails safely; a Gate succeeds WITH human help -- it's planned in advance, not an error.

Where This Leads / Sources

Leads to: Loop Engineering, Harness Engineering, Graph Engineering, Trusting the Checker. Full 8-case diagnostic drill in 03-practice.

Source: docs/four-layers/ (this repo), content originally via Zia Tutor AI connector.